

Spectrum :

An instrument used to separate the radiation of different wavelengths (or frequencies) is called spectroscopy or a spectrograph. Photograph (or the pattern) of the emergent radiation recorded on the film is called a spectrogram or simply a spectrum.

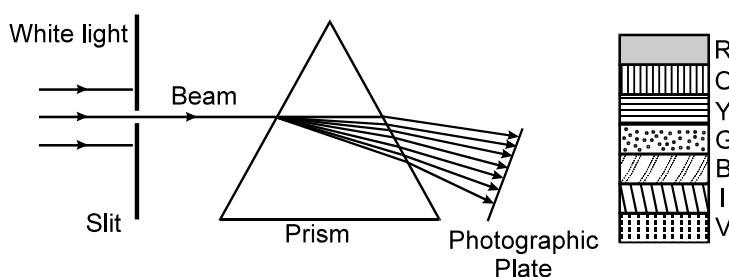
Emission spectra :

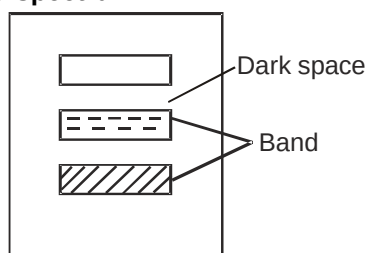
When the radiation emitted from some source e.g. from the sun or by passing electric discharge through a gas at low pressure or by heating some substance to high temperature etc, is passed directly through the prism and then received on the photographic plate, the spectrum obtained is called 'Emission spectrum'.

Depending upon the source of radiation, the emission spectra are mainly of two type :

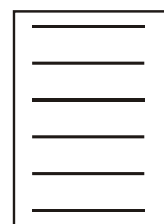
(a) Continuous spectra :

When white light from any source is analysed by passing through a prism, it is observed that it splits up into seven different wide band of colours from violet to red. These colours are so continuous that each of them merges into the next. Hence the spectrum is called continuous spectrum.


(b) Discrete spectra : It is of two type

(i) Band spectrum


Band spectrum contains colourful continuous lines
 bands separated by some dark space

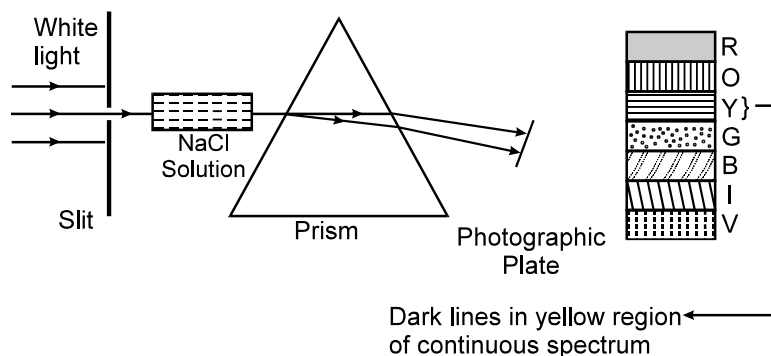
(ii) Line Spectrum :


This is the ordered arrangement of
 of particular wavelength separated by
 dark space eg. hydrogen spectrum.

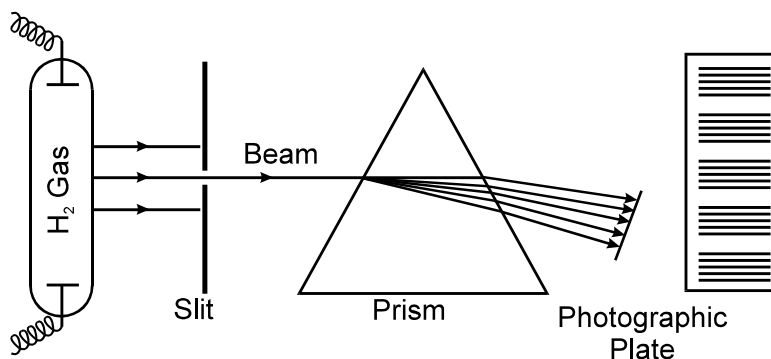
* Generally molecular spectrum are band spectrum while line spectrum can be obtained from atoms.

2. Absorption spectra :

When white light from any source is first passed through the solution or vapours of a chemical substance and then analysed by the spectroscopy, it is observed that some dark lines are obtained in the continuous spectrum. These dark lines are supposed to result from the fact that when white light is passed through the chemical substance, radiations of certain wavelengths are absorbed, depending upon the nature of the element.



EMISSION SPECTRUM OF HYDROGEN :

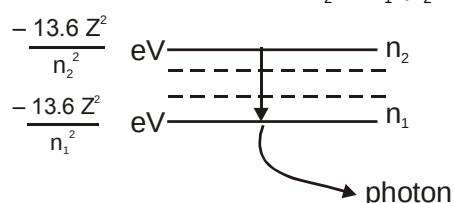


When hydrogen gas at low pressure is taken in the discharge tube and the light emitted on passing electric discharge is examined with a spectroscope, the spectrum obtained is called the emission spectrum of hydrogen.

Line Spectrum of Hydrogen :

Line spectrum of hydrogen is observed due to excitation or de-excitation of electron from one stationary orbit to another stationary orbit

Let electron make transition from n_2 to n_1 ($n_2 > n_1$) in a H-like sample



$$\text{Energy of emitted photon} = (\Delta E)_{n_2 \rightarrow n_1} = -\frac{13.6 Z^2}{n_2^2} - \left(-\frac{13.6 Z^2}{n_1^2} \right) = 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Wavelength of emitted photon

$$\lambda = \frac{hc}{(\Delta E)_{n_2 \rightarrow n_1}} \Rightarrow \lambda = \frac{hc}{13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)} \Rightarrow \frac{1}{\lambda} = \frac{13.6 Z^2}{hc} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\text{Wave number, } \frac{1}{\lambda} = \bar{\nu} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$R = \text{Rydberg constant} = 1.09678 \times 10^7 \text{ m}^{-1}; R \simeq 1.1 \times 10^7 \text{ m}^{-1}; R = \frac{13.6 \text{ eV}}{hc}; R h c = 13.6 \text{ eV}$$